

	Document Type:	Document Code	RES- PROP-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	November 25, 2019
	RESEARCH PROPOSAL	Page	Page 1 of 8

(1) Research Program/Project/Study Title:

Development of Calabash Fruit (*Crescentia cujete*) and Lemongrass (*Cymbopogon Citratus*) enriched with Calamansi (*Citrus microcarpa*) Extra and Ginger (*Zingiber officinale*) Juice Drink

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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component 1 Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ▪ Food Research ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ▪ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation: <table border="1"> <tr> <td>Municipality/City:</td> <td>Roxas City</td> </tr> <tr> <td>Province:</td> <td>Capiz</td> </tr> <tr> <td>Region:</td> <td>VI</td> </tr> </table>				Municipality/City:	Roxas City	Province:	Capiz	Region:	VI
Municipality/City:	Roxas City								
Province:	Capiz								
Region:	VI								
(5) Classification <table border="1"> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td> ☐ Basic ☐ Applied </td> <td> ☐ Pilot Testing ☐ Tech. Promotion/Commercialization </td> </tr> </table>		Research:	Development:	☐ Basic ☐ Applied	☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ▪ Collaborative Research (within campus) ☐ Collaborative Research (intercampus)			
Research:	Development:								
☐ Basic ☐ Applied	☐ Pilot Testing ☐ Tech. Promotion/Commercialization								

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 25, 2019
			Page	Page 2 of 8

		☐ Collaborative Research (external agency)
(7) Sector/Commodity NUTRITIOUS JUICE		(8) Discipline/Field of Specialization Food and Technology
(9) Rationale Global public health is facing enormous challenges as a result of the sharp increase in the burden of chronic health issues such as obesity, cardiovascular disease, and type 2 diabetes in recent decades (World Health Organization, 2021). Amid this health crisis, eating a nutrient-rich diet has become more important as a preventative and treatment strategy (American Heart Association, 2020). In a time when the diet is the most critical factor in overall health and lifestyle, diseases are becoming more prevalent, and eating foods high in nutrients is becoming increasingly important. Among these, drinking nutritious juices has become a well-liked and practical way to increase daily vitamin intake and enhance wellness. Healthy juices from various fruits and vegetables have gained popularity due to their potential to lower the risk factors connected with common health disorders. These beverages have a high concentration of critical vitamins, minerals, antioxidants, and phytochemicals, essential in sustaining physiological processes and lowering disease susceptibility. With its rich biodiversity and dynamic culinary tradition, the Philippines has a broad range of indigenous ingredients known for their distinct tastes and healing benefits. Among them, Calabash fruit (<i>Crescentia cujete</i>), lemongrass (<i>Cymbopogon citratus</i>), calamansi (<i>Citrus microcarpa</i>), and ginger (<i>Zingiber officinale</i>) are notable for their outstanding nutritional profiles and traditional therapeutic purposes. Calabash fruit (<i>Crescentia cujete</i>), sometimes called miracle fruit, is packed with calcium, iron, sodium, potassium, vitamin B, and vitamin C. It has been demonstrated to help with a wide range of ailments, including lowering blood pressure and sugar levels, inhibiting the formation of cancerous cells, effectively treating respiratory issues, including asthma and cough, and preventing the development of cancer cells in the human body (Tacio, 2015). A small citrus fruit called calamansi, "<i>Citrus microcarpa</i>", often called calamondin or Philippine lime, is widely grown in the Philippines and other Southeast Asian nations. Due to its sour and acidic flavor, it is a common component in Filipino and different Southeast Asian cuisines. Calamansi is a good source of vitamin C, which is necessary for a healthy immune system and skin, just like other citrus fruits. Antioxidants and other beneficial ingredients are also present. Calamansi has been used in traditional medicine for some ailments, including coughs, colds, and sore throats. The immune system is supposed to benefit from its high vitamin C concentration. Ginger (<i>Zingiber officinale</i>) is one of the world's most commonly consumed dietary condiments (Surh et al., 1999). Ginger contains a rich array of bioactive compounds, including gingerol, shogaol, and paradol, contributing to its pharmacological properties (Marx et al., 2020). These compounds exhibit potent antioxidant, anti-inflammatory, antimicrobial, and anticancer activities, rendering ginger a versatile remedy for various health conditions (Mashhadi et al., 2013; Hajimonfarednejad et al., 2019). This study aims to provide a nutritious juice utilized by Calabash fruit, Lemongrass, Calamansi, and ginger to create an additional product, extend its shelf life, and increase the production of Calabash fruit, Lemon grass, Calamansi, and ginger in the province of Capiz. It aims to promote long-term usage, cultural preservation, and contribution to public health and well-being in the Philippines and internationally by increasing awareness of these indigenous components.		

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 25, 2019
			Page	Page 3 of 8

(9-A) Project Description *(Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other work needed to accomplish the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)*

The following ingredients comprise the healthy Juice that will be evaluated and analyzed by the research: the primary ingredients are calabash fruit, lemongrass, calamansi extract, and ginger, which are abundant in nutritional elements and readily available locally. Other ingredients are water, agar, and refined sugar. The Juice's acceptance will be determined by its sensory attributes, including color, texture, taste, and odor. Additionally, the acceptability of Nutritious Juice by consumers will be assessed.

Completely Randomized Design will be used with a single factorial design with four (4) treatments and three (3) replications. Analysis of Variance will be the statistical tool to use.

This five-month project will involve material preparation, product development and evaluation, statistical analysis, consumer acceptability, and write-ups at the FFRDC laboratory.

(10) Objectives/Statement of the Problem

The research focus is to determine the development and acceptability of Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger in making healthy Juice Drink

1. To determine the most acceptability of Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger Juice Drink
2. To evaluate the sensory properties of Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger Juice Drink
3. To assess consumer acceptability of the optimized product;
4. To determine the Proximate Analysis of Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger Juice Drink
5. To determine the self-life of Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger Juice Drink

(11) Review of Literature/Prior Art Search *(Related researchers/activities that have been conducted. State-of-the-art of current technology/ information from which the project proposal will take off).*

PH22018001137U1 A Process of Making Calabash (Crescentia cujete) Fruit Juice Enriched with Calamansi Extract and Honey

A process of producing calabash (Crescentia cujete) fruit juice enriched with calamansi (Citrofortunella microcarpa) extract and honey, comprising the steps of: stocking harvested calabash fruits; washing the calabash; cutting the fruits using a saw; scraping the flesh finely using a spoon; cooking the scraped flesh at 75 to 80°C for 20 minutes; cooling the cooked flesh at room temperature; squeezing the cooked flesh using sanitized net bag to separate the juice from the flesh; filtering the juice using a cheese cloth to remove remaining residue; adding 20 pcnt pure calamansi extract and 5 pcnt honey into the calabash juice; placing the mixture in sanitized bottled container; and chilling the bottled calabash juice in a refrigerator.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 25, 2019
			Page	Page 4 of 8

CN102424793B Miracle fruit juice liquor and preparation method thereof

The invention relates to a Miracle fruit juice liquor and a preparation method thereof. The Miracle fruit juice liquor which is prepared through alcoholizing a Miracle fruit juice by a liquor, clarifying, filtering, and blending the obtained liquid with the liquor, sugar and softening water comprises, by weight, 15-30% of the Miracle fruit juice, 40-50% of the liquor, 6-10% of sugar, and the balance softening water. The Miracle fruit juice liquor of the invention, which has the advantages of stable flavor and quality, heavy unique fruit fragrance of the Miracle fruit and liquor fragrance, soft and pure taste, no addition of artificial chemical additives of antiseptics, pigments and the like, environmental protection, health, and complete reservation of original nutritive components of the Miracle fruit, is a nutrition health drink, and the production and the processing of the Miracle fruit juice liquor are no longer limited by the raw material seasonality, so the commercial value of the Miracle fruit is improved.

PH22018000251U1 Lemon Grass Beverage Enhanced with Calamansi

A natural and nutritious health drink particularly a lemongrass juice enhanced with calamansi comprising the following ingredients: lemongrass juice- 240ml; calamansi juice- 20 grams; and syrup - 20 grams.

CN101999720A Method for processing high-pulp content and concentrated lime juice.

The invention belongs to the field of orange juice processing and particularly relates to a method for processing high-pulp content and concentrated lime juice. The method comprises the following steps of: 1) soaking lime fruits into potassium permanganate solution for treatment, spraying chloridized water onto the surfaces of the fruits for sterilization, and spraying the fruits with clean water to obtain clean fruits; 2) squeezing the fruits by using a centrifugal juicer, and collecting lime juice and peel residues; 3) slicing the lime peel residues; 4) debitterizing the lime juice by using macroporous resin HDP800, and debitterizing lime peel in a water soaking mode; 5) adding sugar into the lime peel residues, and curing, freezing and storing the lime peel residues; 6) adding honey into the frozen and stored lime peel residues, concentrating the lime juice, vitamin C, citric acid, golden pomelo essence and the like until the heating temperature is 100 DEG C to obtain semi-finished products of the concentrated lime juice; and 7) sterilizing the lime juice at the temperature of 85 DEG C for 15 minutes to obtain a high-pulp content and concentrated lime juice product. By the method, the flavor substances in the high-pulp content lime juice are preserved to the greatest extent. The method has important significance to the improvement on added value of quince processing industry.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

INPUT	PROCESS	OUTPUT
Calabash Fruit Juice and other ingredients	- Product Preparation and formulation of Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger Juice Drink - Consumer Testing and sensory evaluation - Shelf-life testing	- Optimized formulation of Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger Juice Drink Shelf-stable product - Proximate Analysis of best formulation

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 25, 2019
			Page	Page 5 of 8

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

The study will be experimental research. It will focus on Development of **Calabash Fruit** (*Crescentia cujete*) and **Lemongrass** (*Cymbopogon Citratus*) enriched with **Calamansi** (*Citrus microcarpa*) **Extra and Ginger** (*Zingiber officinale*) **Juice Drink**.

Table 1. Treatments of Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger Juice Drink

Treatment A	Treatment B	Treatment C	Treatment D
100% of Calabash Fruit Juice	75% of Calabash Fruit Juice	50% of Calabash Fruit Juice	25% of Calabash Fruit Juice
NO Lemongrass	25% of Lemongrass	50% of Lemongrass	75% of Lemongrass

Materials and Ingredients

The materials to be use in this study are Knife, Chopping Board, Casserole, Bowl, Stove, Steamer, Ladle, Measuring cup and Spoon, Strainer, Cheese Cloth, Blender, Plastic Bottles and cooking equipment. The ingredients to be use are Calabash Fruit, Lemongrass, Calamansi Juice Extract, Water, Ginger, Refine Sugar, and Agar.

Procedures

Preparation of Sanitizing Plastic Bottles

1. Wash the Bottles and Cups.
2. Soak the Plastic Bottles and Cups in hot water for 5 minutes.

Preparation of Sugar Syrup

1. Dissolve 150g sugar in 80 ml water over low heat.
2. Cool down at room temperature.

Preparation of Calamansi Extract

1. Wash the Calamansi Fruit.
2. Cut in half the Calamansi Fruit.
3. Squeeze the Calamansi using a citrus squeezer or your hands.
4. Strain the squeezed Calamansi Juice to separate the seed from the Juice.

Preparation of Lemongrass Juice

1. Wash the Lemongrass with clean water.
2. Cut the Lemongrass into 2 inches long.
3. Put the Lemongrass and water in the casserole and boil.
4. Cool down at room temperature.

Preparation of Calabash Fruit Juice

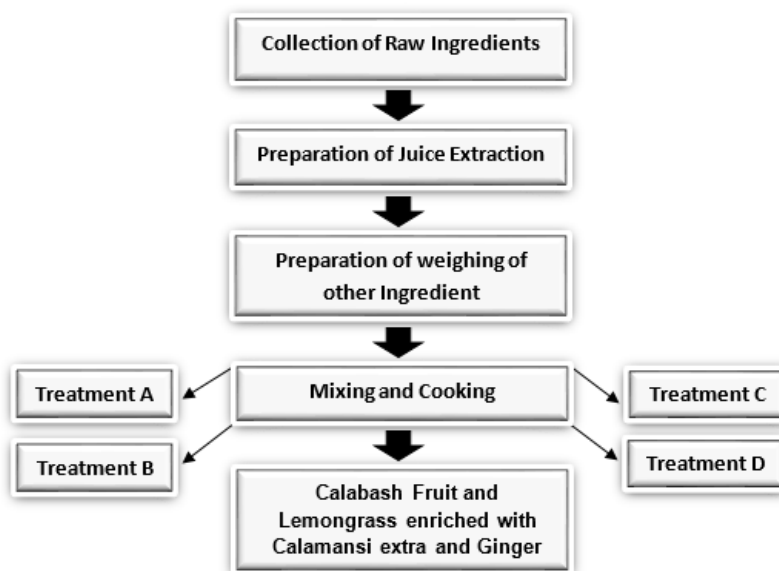
1. Wash Calabash fruit in clean running water.
2. Break the fruit into half using a sharp butcher knife.
3. Scoop all the fruit meat from the shell using a slotted ladle.
4. Mash the Calabash pulp using a masher.
5. Mix the pulp with water.
6. Cook the mixture for 30-45 minutes.
7. Strain the cooked mixture to separate the pulp from the juice.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 25, 2019
			Page	Page 6 of 8

Procedures of Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger Juice Drink

1. Prepare all the ingredients.
2. Put all the ingredients together in a blender (Calabash Fruit Juice, Lemongrass Juice, Calamansi Extract, Sugar syrup, and ginger)
3. Blend all the ingredients until it becomes a homogenous mixture.
4. Strain the mixed Juice to separate the ginger pulp.
5. In a casserole, sterilize the Juice by boiling it for 10 minutes.
6. Remove the concentrate from the casserole and hot fill into the glass container to cool down.
7. Transfer the Juice to a plastic bottle, seal it, and refrigerate.

Figure 1. Flow Chart in development of Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger



Experimental Design

This study will use the experimental method or procedure involving the control or manipulation of conditions to study the relative effects of various treatments applied to several samples or the same treatment applied to several samples. A completely randomized block design (CRD) will be employed during the treatment and sensory evaluation.

Proximate analysis

The Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger Juice Drink will be brought to the DOST laboratory for proximate composition analysis.

Statistical analysis

The data collected on sensory evaluation was statically analyzed by using SPSS software. The analysis of variance (ANOVA) was used to analyze sensory evaluation results with the significant difference among the samples with different treatments.

Data Collection

The data was gathered using a score containing the 9-point hedonic scale. It includes the panelist's color, taste, flavor, texture, and general acceptability rating. The data was gathered after

	Document Type:	Document Code	RES- PROP-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	November 25, 2019
	RESEARCH PROPOSAL	Page	Page 7 of 8

every replication and was collected, tallied, tabulated, and computed using mean for acceptability and Analysis of Variance (ANOVA) for the significant difference among treatments.

Sensory evaluation

The Calabash Fruit and Lemongrass enriched with Calamansi extra and Ginger Juice Drink will be prepared per the sensory evaluation protocol.

Scorecards that contain the 5-point scale for sensory characteristics and the 9-point Hedonic Scale for acceptability will be used to evaluate the development of the product, and 30 evaluators will assess the product.

Only sensory characteristics such as color, taste, flavor, texture, and general acceptability will be evaluated.

For the consumer's preference, 100 respondents of varying ages and sex will evaluate the product.

Scoring Variables

The Criteria for evaluation of the product's sensory characteristics used a 5-point hedonic scale in terms of color, taste, flavor, texture, and general acceptability. The rating for general acceptability in 9- point hedonic scale is 9 for like extremely, 8 for like very much, 7 for like moderately, 6 for like slightly, 5 for neither like nor dislike, 4 for dislike moderately, 3 for dislike slightly, 2 for dislike very much and 1 for dislike extremely (Calmorin 2006).

The score on each item will be interpreted based on the result of the computation regarding color, taste, flavor, texture, and general acceptability using the following scoring interval and verbal interpretation.

General Acceptability

Scoring	Score Interval	Verbal Interpretation/Response
9	8.12-9.0	Like Extremely
8	7.23-8.11	Like Very Much
7	6.34-7.22	Like Moderately
6	5.45-6.33	Like Slightly
5	4.56-5.44	Neither Like nor Dislike
4	3.67-4.55	Dislike Slightly
3	2.78-3.66	Dislike Moderately
2	1.89-2.77	Dislike Very Much
1	1.00-1.88	Dislike Extremely

(14) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Research Objectives	Duration (months)				
	1	2	3	4	5
1. Preparation and selection of Raw Materials					
2. Product Development					
3. Product Evaluation					
4. Shelf life Testing					
5. Statistical Analyses					
6. Manuscript preparation and write-up					

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 25, 2019
			Page	Page 8 of 8

(15) Expected Output (Indicate the specific products, processes or services which the project aims expected to produce; quantify when possible)

1. Utility Model
2. Patent Product
3. IEC material
4. Commercialization Product

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Training on the process of Calabash Fruit (*Crescentia cujete*) and Lemongrass (*Cymbopogon Citratus*) enriched with Calamansi (*Citrus microcarpa*) Extra and Ginger (*Zingiber officinale*) Juice Drink

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

1. Calamansi, Calabash, lemongrass and Ginger farmers
2. Entrepreneurs
3. Researchers
4. Consumers
5. Vegetarians

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Stephanie S. Pimentel	Study Leader	Biology
Janine G. Ferrer	Research Assistant	Biology
Vialyn B. Ragalado	Research Assistant	Food Science

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

Ingredients	Quantity	Amount
Calamansi	30 kls	1,000.00
Other Ingredients (calabash, Ginger, Refined Sugar, Water lemongrass)		1,500.00
Gas during product Development.		1300.00
Plastic cups for Sensory and Plastic Gloves	6packs	1000.00
Proximate Analysis		10,000.00
Transportation Allowance		1000.00
Patent		5,000.00
TOTAL		20,800.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Albaladejo Dr Reyma P [Ph], Univ Capiz State [Ph], Ph22018000251u1-2018-08-22. Lemon Grass Beverage Enhanced With Calamansi

(20) Capsule Curriculum Vitae (One-page brief C.V. only)